

Derivation of the Action Principle by application of outcome counting to the many-worlds interpretation of quantum mechanics

David Strayhorn

Dept. of Neurology, Washington University in Saint Louis

660 South Euclid Avenue

Saint Louis, MO, 63110, USA

strayhornd@neuro.wustl.edu

wdstrayh@alumni.princeton.edu

Outcome counting has been proposed as an alternative to the Born rule as the fundamental probability rule of the Everett many-worlds interpretation (MWI). The primary argument in favor of outcome counting is aesthetic: that it is the most intuitively natural probability rule that one might associate with the MWI. This paper presents an additional argument in favor of outcome counting: that it forms a foundation from which the action principle can be derived.

Keywords: many-worlds interpretation, Born rule, outcome counting, action principle, quantum mechanics

Introduction

Everett's many-worlds interpretation (MWI) of quantum mechanics [1] is one of the major formulations of quantum mechanics [2]. Nevertheless, fifty years after initial publication, there are still philosophical (ontological) issues surrounding the MWI that remain problematic, even among its adherents. Some of these philosophic issues arise as a consequence of the assumption of the Born rule¹ in the formalism. To address these issues, several have proposed that the Born rule should be replaced, in the MWI, with an alternate probability rule known variously as the alternate projection postulate [3], egalitarianism [4], or outcome counting [5, 6]. The essence of outcome counting is that all outcomes are attributed an equal probability measure; it is, in a sense, a "democracy of outcomes." It has been argued elsewhere (including by this author [7]) that outcome counting solves many of the ontologic difficulties of the MWI, assuming it can be made into a workable theory. Indeed, several attempts at a workable theory [8, 5, 6] have been proposed. The purpose of this paper will be to demonstrate that outcome counting, applied to the MWI, can establish a basis for the derivation of an action principle. The implication is that these results may provide a useful tool for the continued attempts to build a workable theory. This should give further credence to the elevation of outcome counting to the status of a symmetry principle.

It would be difficult to underestimate the fundamental role

¹The Born rule assigns a "probability measure" $m = |a_n|^2$ to each of the N experimental outcomes associated with a measurement, with a_n being the complex coefficient associated with the n^{th} outcome, $n \in [1, N]$, as calculated by the Schrödinger equation.

played by action principles in theories of the physical world. Loosely speaking, the action principle “is predicated on the notion that nature always minimizes certain important quantities when a physical process takes place” [9]; or as stated succinctly by Maupertuis, “Nature is thrifty in all its actions.” Typically, the “important quantity” to be minimized – the action S – is defined as the integral over a quantity termed a Lagrangian (or Lagrangian density), and so the trick to define the action boils down to defining the Lagrangian (or Lagrangian density). Examples of action principles go back at least as far as the ancient Greeks, when Hero formulated his principle of the shortest path to determine the path taken by light reflected by a mirror. Fermat reformulated this into his principle of least time, deriving not only the law of reflection but also the law of refraction, by changing the action from the *length* of the path of light to its *time* of travel.

Indeed, nearly all of classical physics can be derived from one or another action principle. Classical mechanics is derived from Hamilton’s principle of least action, with the Lagrangian defined as kinetic energy minus potential energy [9]. Maxwell’s equations can be derived using a similar principle of stationary action. The Einstein field equation can be derived from an action principle, utilizing the Einstein-Hilbert action [10]. It is even possible to derive quantum theory itself from an action principle (the “variational formulation” discussed in [2]).

This paper will outline how a generic action principle could arise from the framework of the MWI, in which the Born rule has been replaced with outcome counting. Section 2 will define the “essential framework” of the MWI in terms of a state transition diagram governing the evolution of the state of the observer.

Section 3 discusses the application of outcome counting to the diagrams of Section 2. Finally, Section 4 identifies a pattern in the organization of the state transition diagram that gives rise immediately to an action principle.

Everett described his formulation not as a theory, but as a metatheory: a statement about a theory, but not the theory itself. The same is true of the contents of this paper. Issues that are not addressed in this paper include 1) an explanation for how or why these patterns might arise in the state transition diagram, and 2) the exact expression for the action S . These issues await the implementation of a specific theory that can be cast within the metatheory of this paper.

Framework of the MWI

In the MWI, history is viewed, not as a single world-line, but as a many-branched tree diagram. The precise definition of a “world” and the precise understanding of what the diagram represents can vary. For the purposes of this paper, the tree diagram will be defined as a directed graph (Figures 1 through 5) that is interpreted as a state transition diagram, where “state” (represented by the open circles) refers specifically to the *state of the observer* ψ^o (following the notation of Everett’s original paper). An arrow from one state to another indicates that it is possible for one state to transition to the other state. The existence of more than one (say, N) arrows emanating from a particular state indicates, simply, that (to quote Everett) “the observer state ‘branches’ into a number of different states” [1]. Each of these N observer states is associated with a probability measure m_n indicating the probability of transition from state

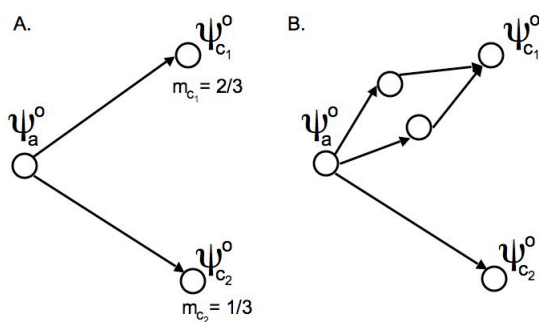


Figure 1: c_1 and c_2 are records of the observables: spin up and spin down, respectively. In part A, the probability measures $2/3$ and $1/3$ are inserted “by hand” by the Born rule. In part B, these same probabilities are derived by outcome counting. The model in part B is trivial, however, in the sense that the state diagram is constructed “by hand” to ensure probabilities that match those provided by the Born rule in part A.

ψ^o to state ψ_n^o . The probability measure is assumed to obey the classical axioms of probability: it is a real number in the interval $[0, 1]$, and the sum over all N probability measures equals 1.

Relative to any given observer-state ψ^o is the associated state of the system under observation ψ^S , where “system under observation” can refer to anything (even to everything!) *external to the observer*. In principle, any physical object can be chosen to play the role of the observer. Typical choices for the observer include a computer, Bob, or Alice. An important characteristic of ψ^o is that it is characterized by an *internal* degree of freedom (x) that is correlated with some *external* degree of freedom (y)

of interest that characterizes ψ^S . For example, y may represent the spin state of an electron, and x may represent a series of bits in computer memory used to record the results of a spin measurement.

In more general terms, define x as a *record* of the *observable* y if there is a correlation between the value of x in the observer state ψ_x^o with the value of y in the associated relative state, ψ_y^S . It is important that there be values of x (bits in the computer) that indicate a record of experimental setup prior to measurement. Additionally, there should be values of x that correspond to incomplete or partial measurements of the observable; in other words, measurements that narrow the value of y down to some subset of possibilities, but do not indicate its exact value.

For instance, consider an experiment involving a spin measurement, yielding two possible spin results $y = \text{“spin up”}$ or $y = \text{“spin down”}$ with the observer being a computer programmed to record all salient aspects of the experiment. This experiment is represented in Figure 1A, in which ψ_x^o represents the state of the computer. $x = a$ is a record of experimental setup prior to measurement, while $x = c_1$ and $x = c_2$ are records of spin up and spin down, respectively.

Outcome Counting

Having drawn the state transition diagram of Figure 1A, the next question becomes: what does the diagram tell us about the *probability* that the observer state ψ_a^o will transition to state $\psi_{c_1}^o$ (or $\psi_{c_2}^o$)? The answer is that the diagram, in and of itself, does not tell us – not in the original Everett interpretation, that

is. Rather, one must (as did Everett) insert “by hand,” so to speak, the probabilities m associated with each distinct branch. (In the example of Figure 1A, these probabilities are $2/3$ and $1/3$ for spin up and spin down, respectively.) In the original Everett interpretation, this is achieved by the *assumption* that the Born rule tells us those probabilities.

Various attempts [1, 11, 12] have been made to demonstrate that the Born rule follows as a necessary consequence of the essential structure of the MWI (with the previous section being a quick summary of the “essential structure” – basically, everything involved in Everett’s reasoning short of assuming the Born rule). Others [13, 3], however, have criticised derivations of these types on the grounds that they contain hidden probabilistic assumptions that effectively “sneak” the Born rule into the formalism.

The obvious alternative to the Born rule was recognized long ago by authors such as Graham, who stated: “it is extremely difficult to see what significance such a measure [the Born rule measure] can have when its implications are completely contradicted by a simple count of the worlds involved, worlds that Everett’s own work assures us must all be on the same footing” [14]. This alternative rule – variously referred to as the alternative probability postulate (APP) [3], egalitarianism [4], or outcome counting [5, 6] – can be stated as follows: given a state ψ_a^o with N possible evolutions $\psi_n^o, n \in [1, N]$, the probability of evolution from ψ_a^o to ψ_n^o is simply $1/N$. This rule is transitive, as illustrated in Figure 2.

There are various ways by which such a scheme can be manipulated to yield correct predictions: that is, a scheme in which outcome counting is the fundamental probability rule at the fine-

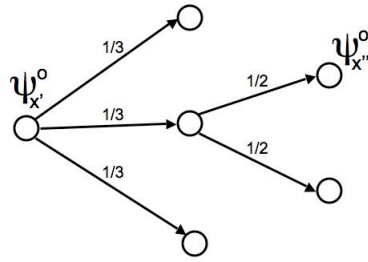


Figure 2: The outcome rule is transitive, so that the probability for an observer in state $\psi_{x'}^o$ to transition to state $\psi_{x''}^o$ is $\frac{1}{3} * \frac{1}{2} = \frac{1}{6}$.

grained level, with the Born rule somehow “emerging” at the coarse-grained level. Such ways will not be the subject of this paper. To demonstrate that it is at least trivially *possible* to do this, consider the following model – illustrated in Figure 1B – in which the number of branches associated with a given experimental outcome is assumed to be proportional to the Born-rule-derived probability of that outcome. Non-trivial models are developed elsewhere [8, 5, 6].

Patterns within the state transition diagram that give rise to an action principle

Unlike the case with the Born rule, once outcome counting is assumed as the fundamental probability rule, it becomes possible to calculate probabilities (the probability that the world-line of an observer beginning in state $\psi_{x'}^o$ will go through state $\psi_{x''}^o$) simply by inspection of the state diagram itself. This is illus-

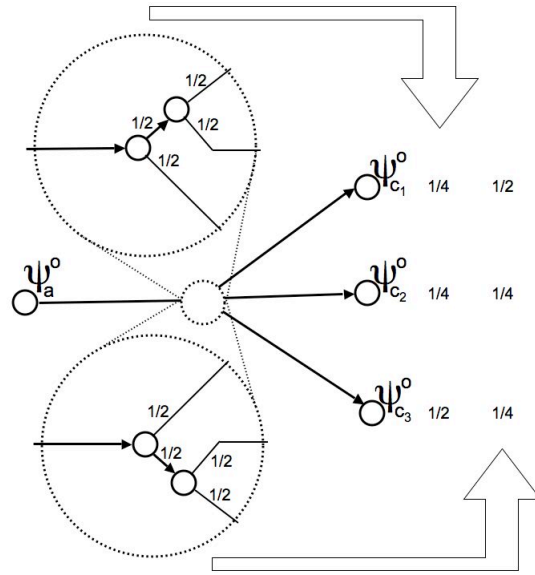


Figure 3: The probability of transition from state ψ_a^o to $\psi_{c_i}^o$, indicated by the arrows, is dependent on the fine scale structure of the state transition diagram.

trated in Figure 2.

An important feature of outcome counting is that the fine scale structure of the state transition diagram can be critical in the calculation of these probabilities, as illustrated in Figure 3. This dependence can be exploited to achieve the main thrust of this paper, which is contained in the observation:

Observation 1 *There exist patterns in the organization of the state transition diagram that result in the minimalization (more generally, extremalization) of an observable.*

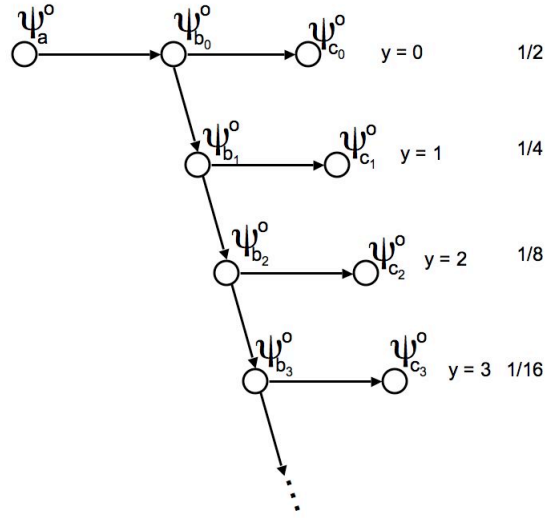


Figure 4: Pattern for the minimization of the observable y .

An example pattern, along with its minimalized observable y , is shown in Figure 4. ψ_x^o represents the state of the observer. As before, $x = a$ indicates a record of experimental setup prior to measurement, $x = b_i$ indicates records of incomplete measurements, and $x = c_y$ indicates a record of the observable y . Assume that y is an integer in $[0, Y]$, with Y some (arbitrarily large) number. Note that $x = b_i$ is a record that the observable y is in $[i, Y]$. Several of the associated probability measures, calculated via outcome counting, are indicated at the right of the figure. Although it is *possible* for the observable y to take any value in $[0, Y]$, it should be evident from the diagram that the

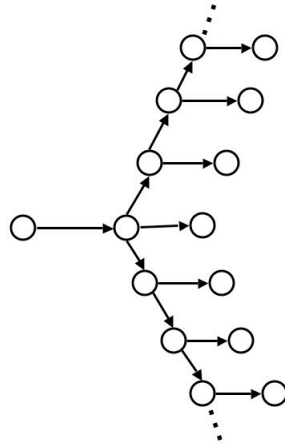


Figure 5: Pattern corresponding to a saddle point.

most *probable* value is its minimum value, $y = 0$.

It is likewise similar to construct a state transition diagram that predicts an observable to take its *maximum* value simply by an appropriate rearrangement of states on the diagram. A saddle point can be achieved by making a bivalent diagram, with the saddle point at the vertex (Figure 5). Therefore, the method here to construct a minimalization principle can be generalized to that of an extremal (maximal, minimal, or saddle point) principle.

How, or why, might this sort of pattern exist in the state transition diagram? This question is to be answered by an underlying theory of observer states – a theory that is not provided by this paper. Nevertheless, the following analogy may help stir the imagination.

Suppose that Alice is driving down a long road and begins to see the tip of a mountain at the horizon. She decides to estimate the mass y of the mountain, which she can do visually once she can see all of it. As she drives down the road, more and more of the mountain becomes visible over the horizon, until finally the bottom of the mountain becomes visible, at which time she records her estimate of y . The state transition diagram – which will resemble the one in Figure 4 – can be constructed by dividing the road into segments and defining the state of Alice at each of these segments. ψ_x^o represents Alice, with x her internal record (thought, in this case) of y . As above, a is the thought that she has not yet seen the mountain; b_i is the record that she has seen a portion of the mountain over the horizon, so its mass is equal to or greater than i ; and c_i is her thought that she has seen the entire mountain from tip to base and so can estimate its mass to be i . Note that whenever Alice is in state $\psi_{b_i}^o$, the next segment of road will yield one of two possibilities: either she sees the base of the mountain and transitions to state $\psi_{c_i}^o$, or she sees more mountain come over the horizon, the base not yet visible, and transitions to state $\psi_{b_{i+1}}^o$. If outcome counting is applied to this state transition diagram, then the prediction is that mountains, in general, can be of any mass, but are much more likely to be of the least possible mass (0).

Suppose now that the underlying theory is one (such as special relativity) that places a “speed limit” on the transmittal of information. Suppose further that the observable y is not the mass of a mountain, but rather a more abstract object with the characteristics of an action: the integral of a suitably chosen lagrangian density in the interior of a 4-dimensional region Ω . The part of Ω nearest the observer is like the tip of the moun-

tain, and the part of Ω farthest from the observer is like the base of the mountain. Just as Alice “sums up” the mass of the mountain a piece at a time, so too does the more abstract observer “integrate over” the volume of Ω a piece at a time, from the nearest to the farthest segment. Just as the pattern of the diagram in Figure 4 acts to minimize the mass of the mountain, so too does it act to minimize the action.

Discussion

The purpose of this paper has been to outline a method for the derivation of an action principle from the essential framework of the many-worlds interpretation based upon two assumptions: 1) that probability is determined by outcome counting rather than the Born rule; and 2) that a class of patterns similar to that of Figure 4 is inherent to the makeup of the state transition diagram. The presentation of this paper is intended as a heuristic argument of feasibility rather than as a rigorous proof. It is envisioned [6] that the Born rule will emerge – perhaps by derivation from an action principle – at the coarse-grained level, with probability governed at the fine grained level by the more ontologically palatable outcome counting.

The motivation behind the first assumption is that, from a purely ontologic perspective, it seems clear that outcome counting is preferable to the Born rule. Of course, theory cannot be dictated purely by ontologic considerations, and the appeal of the Born rule over outcome counting is simple and undeniable: *it works*. This appeal is so strong that the ontologic advantages of outcome counting can be easily forgotten, overlooked, or underemphasized. Indeed, considerable effort has been invested

in the construction of arguments to the effect that the Born rule is better than unproblematic: that it is, in fact, *inevitable*. Nevertheless, there is a growing literature dedicated to the unresolved philosophical and ontological difficulties surrounding the probability interpretation of the MWI. The occurrence of two international conferences focused on the probability interpretation of the MWI, set to occur this year at the fiftieth anniversary of Everett's original paper, suggests that this issue is a timely one.

Relatively little effort, on the other hand, has been invested in the converse approach: the search for a workable theory based on outcome counting. It is hoped that the prospect of deriving an action principle from outcome counting will provide a tool in the construction of such a theory.² Such a theory must provide a detailed mathematical description of the set of all possible observer-states, accompanied by a detailed mathematical description of the state transition diagram. In addition, it must lay out how to calculate a mathematical description of the system ψ^S relative to ψ^o . This must include a specification of internal degrees of freedom y and x , including when and why x meets the criteria (above) to be a record of y . The derivation and implementation of a specific action principle then depends upon the demonstration of the existence of the class of patterns in the state transition diagram as discussed above.³ Such a theory is not provided in this paper, but is the topic of future work.

²An example of how this tool may be utilized is provided in [6].

³Whether the transition from one state to another in the state diagram remains discrete, as is the case in this paper, or whether a continuous flow develops in some sort of limit, will be dependent upon the specific underlying theory.

References

- [1] Hugh Everett. Relative state formulation of quantum mechanics. *Reviews of Modern Physics*, 29:454–462, 1957.
- [2] Daniel F Styer, Miranda S. Balkin, Kathhryn M. Becker, Matthew R. Burns, Christopher E. Dudley, Scott F. Forth, Jeremy S. Gaumer, Mark A. Kramer, David C. Oertel, Leonard H. Park, Marie T. Rinkoski, Clait T. Smith, and Timothy D. Wotherspoon. Nine formulations of quantum mechanics. *Am. J. Phys.*, 70:288–297, 2002.
- [3] Patrick Van Esch. On the everett programme and the born rule. *quant-ph/0505059*, 2005.
- [4] Hilary Greaves. Understanding deutsch’s probability in a deterministic multiverse. *Studies in the History and Philosophy of Modern Physics*, <http://philsci-archive.pitt.edu/archive/00001742>, 2004.
- [5] M. B. Weissman. Emergent measure-dependent probabilities from modified quantum dynamnics without state-vector reduction. *Foundations of Physics Letters*, 12:407–426, 1999.
- [6] David Strayhorn. Derivation of the born rule from outcome counting and a solution to the quantitative problem of the multiple worlds interpretation. *Philica.com*, Article number 28, 2006.
- [7] David Strayhorn. An illustration of the quantitative problem of the multiple worlds interpretation of quan-

tum mechanics and the motivation for outcome counting. *Philica.com*, Article number 27, 2006.

- [8] Robin D. Hanson. When worlds collide: Quantum probability from observer selection? *Foundations of Physics*, 33:1129 – 1150, 2003.
- [9] Jerry B. Marion and Stephen T. Thornton. *Classical Dynamics of Particles and Systems*. Harcourt Brace Jovanovich, Inc., Orlando, Florida, third edition, 1988.
- [10] Charles W. Misner, Kip S. Thorne, and John Archibald Wheeler. *Gravitation*. W. H. Freeman and Company, New York, New York, 1973.
- [11] JB Hartle. Quantum mechanics of individual systems. *American Journal of Physics*, 36:704–712, 1968.
- [12] David Deutsch. Quantum theory of probability and decisions. *Proc. R. Soc. Lond. A*, 455:3129–3137, 1999.
- [13] H. Barnum, C. M. Caves, J. Finkelstein, C. A. Fuchs, and R. Schack. Quantum probability from decision theory? *Proc. Roy. Soc. Lond. A*, 456:1175–1182, 2000.
- [14] Neill Graham. The measurement of relative frequency. *The Many-Worlds Interpretation of Quantum Mechanics*, pages 229–252, 1973.